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OpenCaseLaw: A Large-Scale Open Corpus and Citation Graph of Published Swiss Court Decisions

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Abstract

Switzerland publishes court decisions across a fragmented infrastructure of 26 cantonal portals, multiple federal court websites, and regulatory agencies. No unified open resource exists. We release *OpenCaseLaw*, a large-scale open corpus of published Swiss court decisions together with a resolved citation and statute-reference graph. The April 10, 2026 snapshot contains 965,141 full-text decisions from 104 courts and regulatory bodies across all 26 cantons and the period 1875–2026, in German (46.6%), French (45.1%), and Italian (8.3%). The accompanying reference graph contains 6.53 million resolved decision-to-decision citation links and 11.34 million decision-to-statute references, with both endpoints navigable: 80 federal laws (39,000 articles) are mirrored locally from Fedlex and approximately 33,000 additional federal and cantonal texts are queryable through a cached integration with LexFind.ch. To our knowledge, this is the first open dataset to couple comprehensive cantonal and federal coverage with a resolved citation graph and dereferenceable statute references for Swiss case law.

We report one empirical finding: a naive reading of the raw cross-language citation matrix appears to show strong German dominance of French and Italian jurisprudence, but this is an artefact of how the *Bundesgerichtsentscheide* (BGE) leading-case collection is distributed by language. 71.6% of BGE records in the corpus are in German, because BGE records inherit the language of the underlying Federal Supreme Court proceedings, which in turn mirrors the Swiss population mix. Once this confound is controlled for — either by excluding BGE targets, or by normalising BGE citations against the BGE language base rate — each language group exhibits a clear preference for its own-language sources. French courts cite French non-BGE sources 4.6 times more often than

German non-BGE sources, and within BGE targets, French courts over-cite French BGEs by 13.5 percentage points relative to the base rate — the largest own-language preference in the matrix.

The dataset is released as Parquet on Hugging Face under CC0-1.0; all code is MIT-licensed. A reproducible snapshot, a 100-query multilingual retrieval benchmark, and the reference-graph database are pinned to a specific commit for the version accompanying this paper.

Keywords: legal corpus, Swiss law, case law, citation network, multilingual, open data, legal information retrieval

1. Introduction

Switzerland publishes court decisions through a fragmented infrastructure: 26 cantonal portals, multiple federal court websites, and various regulatory agencies. No unified public interface exists. Commercial aggregators such as Swisslex and Weblaw provide broader coverage but require paid subscriptions and prohibit bulk access. Existing open datasets are limited to a single court (typically the Federal Supreme Court) or to narrow NLP benchmarks.

This paper describes *OpenCaseLaw*, an open corpus of published Swiss court decisions with a resolved citation graph. We make three concrete contributions:

1. **A large-scale open corpus** of 965,141 published decisions from 104 courts and regulatory bodies, with full text in German, French, and Italian. To our knowledge, this is the largest open collection of published Swiss court decisions and the only one that combines comprehensive cantonal and federal coverage with trilingual full text.
2. **A resolved citation and statute-reference graph, plus a legislation access layer.** We extract and resolve 8.87 million citation references (73.6% resolved to in-corpus targets) and 11.34 million decision-to-statute links across 284,145 distinct provisions. We release the graph as a SQLite database alongside the corpus. Critically, the statute-reference endpoints on each citation are navigable: 80 Swiss federal laws (39,000 articles) are mirrored locally from Fedlex (the official Swiss federal legislation portal) and served via a fast lookup interface, and approximately 33,000 additional federal, cantonal, and intercantonal legislative texts are queryable through a cached integration with LexFind.ch. This bridges a practical gap for LLM-based legal research, where the Fedlex SPARQL endpoint and the LexFind multi-step search API are difficult to access directly; LLMs asked about Swiss statutes without this layer typically hallucinate article text rather than retrieve the authoritative current version.
3. **A reproducible retrieval baseline.** We release a 100-query multilingual benchmark with graded relevance judgments and report results for a BM25 + citation-graph pipeline. All artifacts are pinned to specific commit hashes for reproducibility.

Scope and terminology. Throughout this paper, “published” means decisions made publicly available by the originating court through an online publication channel. Unpublished decisions—the majority of Swiss court output—are out of scope. The corpus reflects what courts choose to publish; it does not claim to be complete with respect to all decisions ever rendered. We are careful to state empirical claims relatively (“to our knowledge”) rather than absolutely, given that commercial database sizes are not publicly audited.

The remainder of the paper describes the corpus (Section 3), the citation and statute reference graph

with a deliberately conservative cross-language analysis (Section 4), a data-quality and validation section (Section 5), a reproducible retrieval baseline (Section 6), availability and reproducibility (Section 7), and ethical considerations and limitations (Section 8).

2. Related Work

2.1 Swiss Legal Datasets

Several open datasets have been released for Swiss legal NLP, but all are limited in scope.

Table 1. Swiss case law resources. Dataset citations: Swiss-Judgment-Prediction (Niklaus et al., 2021), SCD (Geering & Merane, 2024), SCALE (Rasiah et al., 2023), MultiLegalPile (Niklaus et al., 2023). Commercial sizes are self-reported estimates; exact figures are not publicly audited. “Graph” indicates whether a citation graph is provided at release.

Resource	Decisions	Courts	Graph	Full text	Access	Updated
OpenCaseLaw (this work)	965,141	104	6.53M	100%	CC0	Daily
Swiss-Judgment-Prediction	85,000	1	No	Partial	Open	No
SCD	122,000	1	No	Yes	Open	No
SCALE	—	1	No	Partial	Open	No
MultiLegalPile (Swiss subset)	n/a	Mixed	No	Yes	Open	No
Swisslex (commercial, est.)	~500k+	Many	No	Yes	Paid	Yes
Weblaw (commercial, est.)	~735k+	Many	No	Yes	Paid	Yes
entscheidsuche.ch	~700k+	Many	No	Yes	Free	Yes

Swiss-Judgment-Prediction (Niklaus et al., 2021) provides 85,000 Federal Supreme Court decisions labeled with binary outcome for judgment prediction. *SCD* (Geering & Merane, 2024) extends this to 122,000 decisions with 31 structured variables. Both are limited to a single court and lack citation structure. *SCALE* (Rasiah et al., 2023) defines benchmark tasks on Swiss legal text without releasing a general-purpose corpus. *MultiLegalPile* (Niklaus et al., 2023) assembles 689 GB of multilingual legal text for language model pretraining; its Swiss subset lacks per-court metadata and citation links. OpenCaseLaw differs in three respects: it covers the full Swiss published court system rather than a single court, it provides a resolved citation graph, and it is continuously updated rather than a static snapshot.

2.2 International Legal Corpora and Citation Networks

The *Caselaw Access Project* (Harvard Law School, 2018) digitized 6.7 million US court decisions—the closest international analogue in ambition. *GerDaLIR* (Althammer et al., 2021) is a German legal information retrieval benchmark with 131,000 court decisions. *COLIEE* (Kano et al., 2024) provides competition tasks for case law retrieval on Canadian and Japanese corpora. *LexGLUE* (Chalkidis et al., 2022) benchmarks legal NLP on English-language tasks.

Citation network analysis of legal corpora has a substantial literature. Fowler et al. (2007) analyse the US Supreme Court citation network to identify influential precedents through network centrality. Bommarito and Katz (2010) study the complexity of the US Code through citation structure. Whalen (2016) examines citation dynamics in Canadian case law. Walzl et al. (2017) apply network analysis to German legal texts. To our knowledge, no prior work has constructed or analysed a citation network at national scale for Swiss jurisprudence.

2.3 Legal Information Retrieval

BM25 (Robertson & Zaragoza, 2009) and Reciprocal Rank Fusion (Cormack et al., 2009) remain strong baselines in legal retrieval. Locke et al. (2024) survey the field and note that domain-specific retrieval systems often outperform general-purpose models on legal text. Shao et al. (2020) demonstrate that incorporating citation signals improves legal case retrieval.

3. Corpus Description

3.1 Collection and Processing

OpenCaseLaw is assembled by 58 court-specific scrapers that download decisions from each court’s publication portal and normalise them into a unified schema. Scrapers run daily via systemd timers on a dedicated server. Each decision is stored as a structured record with 34 fields including court identifier, docket number, decision date, language, legal area, headnote (*Regeste/Résumé*), full text, chamber, and source URL.

Deduplication uses a canonical key derived from normalised court code, docket number, and decision date. Within-court deduplication retains the version with the richest content. Cross-court deduplication operates only within hand-maintained overlap groups (e.g., between a *Bundesgerichtssentscheide* (BGE) leading case and its underlying Federal Supreme Court ruling). BGE leading cases are retained as distinct records because they differ from the underlying BGer ruling in court code, docket number, editorial scope, and regeste; Section 4.1 discusses how this affects graph analysis.

The corpus is exported as 104 Parquet files (one per court, ~7 GB total) and uploaded to Hugging Face. The full pipeline—scraping, deduplication, FTS5 index build, Parquet export, and upload—runs unattended daily.

3.2 Corpus Statistics

Table 2. Corpus summary (April 10, 2026 snapshot).

Metric	Value
Total decisions	965,141
Courts and public bodies	104
Cantons represented	26 of 26
Temporal range	1875–2026
German decisions	449,780 (46.6%)
French decisions	435,611 (45.1%)
Italian decisions	79,750 (8.3%)

Romansh-language decisions are not represented because the four courts operating in Romansh (in Graubünden) do not publish electronically in Romansh.

Temporal distribution. The corpus reflects both the expansion of Swiss judicial publication and the digitization of historical archives. Table 3 buckets every decision in the snapshot, including pre-1870 early federal decisions (796), 2026-year-to-date decisions (6,233), and decisions where a reliable date could not be extracted (4,932). The buckets sum exactly to 965,141.

Table 3. Decisions by decade. Buckets sum to the corpus total.

Period	Decisions	Period	Decisions
pre-1870	796	1950s	2,304
1870s	659	1960s	2,298
1880s	1,042	1970s	4,526
1890s	1,646	1980s	10,545
1900s	2,372	1990s	31,503
1910s	2,281	2000s	189,134
1920s	1,978	2010s	416,026
1930s	2,022	2020s	282,854
1940s	1,990	2026 YTD	6,233
		Date unknown	4,932

Since 2015, the corpus grows by approximately 47,000 decisions per year, reflecting the current publication rate of Swiss courts. The 4,932 decisions with unknown dates are concentrated in older historical sources and scanned-PDF collections where date extraction failed.

Court distribution. Table 4 lists the ten largest sources by decision count. The Vaud cantonal courts appear in three entries because the scraper tracks three distinct publication snapshots (`vd_gerichte` current, `vd_findinfo` 2008+ archive, `vd_omni` pre-2008 archive) that are retained as separate records because the corresponding source URLs differ; combined they contribute 155,639 decisions, which would make Vaud the third-largest source after the Federal Supreme Court and Geneva.

Table 4. Top 10 sources by decision count.

Source	Decisions
Federal Supreme Court (bger)	174,779
Geneva cantonal courts (ge_gerichte)	167,346
Federal Administrative Court (bvger)	91,919
Vaud FindInfo archive (vd_findinfo, 2008+)	74,819
Ticino cantonal courts (ti_gerichte)	59,341
Vaud current (vd_gerichte)	52,788
BGE leading cases (bge)	35,815
Zurich Social Insurance Court	33,761
Vaud Jurisweb archive (vd_omni, pre-2008)	28,032
Zurich Upper Court (zh_obergericht)	27,499

The long tail comprises specialised courts, regulatory bodies (FINMA, WEKO, EDOEB, ELCOM, PostCom, ComCom), historical sources (BStGer since 2004, BPatGer, EMARK asylum decisions), and attorney disciplinary authorities.

3.3 Field Completeness and Missingness

Swiss courts publish decisions with varying levels of structured metadata. Table 5 reports the share of records with a non-empty value for each of the most commonly-used fields. Field completeness

should be interpreted alongside the knowledge that some courts (notably BGer and BVGer) produce rich structured metadata while many cantonal courts publish only full text.

Table 5. Field completeness by optional field.

Field	Non-empty	Share
docket_number	965,115	100.0%
decision_date	960,209	99.5%
regeste (headnote)	562,165	58.2%
title	351,641	36.4%
chamber	300,144	31.1%
decision_type	140,389	14.5%
outcome	90,735	9.4%
full_text	965,141	100.0%

All decisions have full text. Docket number and decision date are near-universal. Headnote, title, chamber, decision type, and outcome are systematically available only for a subset of courts (primarily federal and larger cantonal courts).

3.4 Full-Text Length

Text length distributions vary by language. French decisions tend to be longer on average, likely reflecting the drafting norms of courts in Romandie, particularly the prolific Geneva cantonal judiciary. The 10th-percentile German texts include brief procedural orders and cost rulings that are substantively thin but legally published.

Historical material (pre-1990) contains OCR artifacts in some cases. A targeted repair pass in March 2026 re-extracted full text via PDF parsing for 33,884 short-text entries, but approximately 5,000 scanned-PDF decisions remain candidates for improved OCR.

4. Citation and Statute Reference Graph

The citation graph is the main structural contribution of this work. It transforms the corpus from a collection of independent documents into a network of legal authority that can be queried, visualised, and analysed.

4.1 Extraction and Resolution

Citation extraction uses regular expressions tuned to Swiss legal citation conventions. We extract three reference types:

- **BGE references** (BGE 131 III 115, ATF 140 I 201, DTF 136 V 117): volume, division, and page of the *Amtliche Sammlung / Recueil officiel / Raccolta ufficiale*, in all three language variants.
- **Docket numbers** (4A_372/2019, 2C_1084/2013): Federal Supreme Court and other federal court case numbers.
- **Statute references** (Art. 41 OR, Art. 8 BV, art. 29 Cst.): article, optional paragraph and letter, and statute abbreviation.

Each extracted decision-to-decision citation is resolved against the corpus using normalised docket matching; BGE references are mapped to their corresponding decisions via volume-division-page lookup. Table 6 summarises the resulting graph.

Table 6. Citation graph summary.

Metric	Value
Extracted decision citation references	8,870,000
Resolved in-corpus links	6,533,534
Resolution rate	73.7%
Decision-to-statute links	11,340,907
Distinct statute provisions cited	284,145
Unique citing decisions (source ends)	680,421
Unique cited decisions (target ends)	207,938

The 26.3% of references that do not resolve to an in-corpus target are concentrated in (a) older decisions not covered by online publication, (b) cantonal rulings cited by federal courts but not published by the originating canton, and (c) decisions from foreign jurisdictions (ECtHR, CJEU, German *Bundesgerichtshof*, etc.). We do not treat unresolved references as errors; they reflect the coverage gap between the published and citing corpora rather than extraction failures.

BGE/BGer duplication. BGE leading cases are published separately from the underlying Federal Supreme Court ruling, and both are indexed in the corpus. When a decision cites “BGE 140 III 264” we resolve it to the BGE record, not to the BGer record. In Section 4.3 we analyse BGE-citation patterns separately to avoid double-counting.

4.2 In-Degree Distribution

The in-degree distribution follows the heavy-tailed shape typical of citation networks (Fowler et al., 2007). A small number of “landmark” decisions concentrate a disproportionate share of all incoming citations.

Table 7. In-degree distribution (incoming citation counts per decision).

Citations received	Decisions
1	63,178
2–5	64,554
6–10	25,622
11–50	38,567
51–100	7,406
101–500	6,685
501–1,000	1,026
1,000+	900

The 900 decisions cited more than 1,000 times correspond to the core of Swiss leading-case jurisprudence. Table 8 lists the five most cited decisions; all five are BGE rulings on social insurance law,

reflecting both the volume of social insurance litigation in Switzerland and the high procedural standardisation in this area.

Table 8. Most-cited decisions.

Decision	Subject area	Citations
BGE 125 V 351	Social insurance	61,981
BGE 134 V 231	Social insurance	35,078
BGE 122 V 157	Social insurance	28,174
BGE 130 V 343	Social insurance	22,484
BGE 137 V 210	Social insurance	20,604

4.3 Cross-Language Citation Patterns

Swiss courts write in the official language of their canton or in the language of the underlying proceedings; the Federal Supreme Court produces each decision in one language only. Citations that cross language boundaries are therefore observable in the corpus, and a naive reading of the raw citation matrix would suggest a strong asymmetry — German sources appear to be cited disproportionately by French- and Italian-speaking courts. We show in this section that this naive reading is wrong, and that the apparent asymmetry is almost entirely an artefact of how the *Bundesgerichtsentscheide* (BGE) leading-case collection is distributed across languages.

Table 9a. Raw cross-language citation matrix (all 6,533,534 resolved links). Rows = citing-court language; columns = cited-decision language.

Citing language	Cited: DE	Cited: FR	Cited: IT	Total
German	2,074,160	359,280	24,328	2,457,768
French	1,642,206	1,964,785	51,766	3,658,757
Italian	239,843	116,939	60,227	417,009

On its face, Table 9a suggests that French-speaking courts cite German-language sources 1,642,206 times versus French-language sources 1,964,785 times — an 84% ratio. But this is misleading for two reasons.

The BGE base-rate confound. The Swiss Federal Court publishes each BGE leading case in only one language, corresponding to the language of the underlying proceedings. The distribution of BGE records in our corpus is 71.6% German (25,637 records), 24.4% French (8,723), and 4.1% Italian (1,455), approximately mirroring the Swiss population distribution of Federal Court cases. When any court cites a BGE, the language of the target is determined by the language of the original proceeding, not by the citing court’s preference. Because 72% of BGE records are German by construction, BGE citations will always be predominantly German targets regardless of which court is citing them.

Table 9b removes this confound by excluding citations whose target is a BGE record.

Table 9b. Cross-language citation matrix with **BGE targets excluded** (2,328,745 links, = 6,533,534 total – 4,204,789 BGE targets).

Citing language	Cited: DE	Cited: FR	Cited: IT	Total
German	760,264	89,846	6,151	856,261
French	234,136	1,082,541	12,209	1,328,886
Italian	54,065	40,681	48,852	143,598

Once BGE targets are removed, the apparent asymmetry reverses. French-language courts cite French-language sources **4.6 times** more often than German-language sources (1,082,541 vs 234,136). Italian-language courts cite German and Italian sources at near-parity (54,065 vs 48,852). German-language courts cite German sources almost exclusively (760,264 vs 89,846 to French). In other words, once the Federal Court’s BGE language mix is controlled for, each language group exhibits a clear home-language preference.

Language preference within BGE citations. A sharper test of whether courts prefer their own language asks: among BGE citations, do courts over-select BGE records in their own language relative to the base rate? Table 9c reports each citing language group’s BGE-target distribution, compared with the 71.6% / 24.4% / 4.1% base rate.

Table 9c. BGE-target language distribution per citing-language group, compared with the corpus-wide BGE language base rate. Positive deviations mean the group over-cites BGEs in that language relative to what would be expected if citations were selected uniformly from the BGE collection.

Citing group	BGE-DE cited	BGE-FR cited	BGE-IT cited	Own-language deviation
German courts	82.0% (+10.4pp)	16.8% (−7.6pp)	1.1% (−3.0pp)	+10.4pp toward DE
French courts	60.4% (−11.2pp)	37.9% (+13.5pp)	1.7% (−2.4pp)	+13.5pp toward FR
Italian courts	67.9% (−3.7pp)	27.9% (+3.5pp)	4.2% (+0.1pp)	+0.1pp (at base rate)
Base rate (BGE corpus)	71.6%	24.4%	4.1%	—

French courts over-cite French BGEs by 13.5 percentage points relative to the base rate — the *largest* own-language preference in the matrix, stronger even than the German courts’ 10.4 percentage-point preference for German BGEs. Italian courts show essentially no preference: their BGE citations match the base rate, reflecting that the Italian-language BGE corpus is small and Italian courts must cite across languages by necessity.

Summary. The three views together reverse the naive interpretation:

1. The raw matrix (9a) shows apparent German dominance, but this is confounded by the BGE language base rate.
2. Excluding BGE targets (9b) reveals that French courts cite French sources 4.6× more than German, and Italian courts cite IT and DE at near parity.
3. Within BGE targets (9c), French courts over-cite French BGEs at the *highest* rate of any group, not lowest.

The corpus-level asymmetry in Swiss citation patterns is therefore best described as a Federal Court language-mix effect, not as a cross-language dominance pattern. Each language group strongly

prefers its own-language sources when they exist, and is forced to cite across languages only when own-language material is unavailable (which happens most often for Italian courts, because the Italian-language corpus is smallest).

These observations are descriptive and corpus-level; they do not control for court type, publication volume per court, or temporal effects. We release the graph as a SQLite database so that researchers can perform controlled analyses of legal harmonisation and cross-language authority flow using the methodology of their choice.

4.4 Citation Temporal Decay

Table 10 buckets the 6,494,553 resolved citation links for which both source and target decision dates are known. A further 38,981 resolved citations (0.6%) are excluded from the age analysis because either the source or the target lacks a parseable date. A small number of links have negative age (citing decision dated before the cited decision); these arise primarily from inter-court appeals where our date extraction pulls the appeal-instance date rather than the original ruling date.

Table 10. Citation age (years between cited and citing decisions), resolved links with dates on both ends. Buckets sum to 6,494,553.

Citation age	Links
Negative (source before target)	56,040
Same year (0)	265,185
1–2 years	958,305
3–5 years	1,274,589
6–10 years	1,551,441
11–20 years	1,588,264
21–50 years	753,133
51+ years	47,596
Total	6,494,553

Peak citation age is 11–20 years. Citation volume remains substantial for citations aged 21–50 years, consistent with Swiss courts exhibiting long institutional memory. *Important caveat:* the age distribution is affected by growth in corpus volume over time (very old decisions have had more time to be cited, but the citing base was much smaller in earlier decades) and by changes in publication practice. A proper analysis would need to normalise by the size of the citing base per year. We report the raw distribution and leave the controlled analysis to future work.

4.5 Statute Reference Network

The 11.34 million decision-to-statute links connect the case law corpus to the statutory framework. Swiss statutes are known by multiple abbreviations depending on language (for example, the Federal Court Act appears as *Bundesgerichtsgesetz* / BGG in German, *Loi sur le Tribunal fédéral* / LTF in French, and *Legge sul Tribunale federale* / LTF also in Italian). Our extraction preserves the surface form used in each decision; Table 11 presents both the raw surface-form totals and the canonicalised totals after merging aliases.

Table 11. Most-cited statutes after merging language aliases. Mentions are counted across all article-level provision identifiers sharing the same canonical statute.

Canonical statute	Surface forms merged	Mentions
Federal Court Act	BGG + LTF	1,715,464
Civil Procedure Code	ZPO + CPC	991,614
Criminal Procedure Code	StPO + CPP	905,448
Penal Code	StGB + CP	472,355
Civil Code	ZGB + CC	426,270
Code of Obligations	OR + CO	327,071
Federal Constitution	BV + Cst.	299,540
Foreign Nationals Act	AIG + LEtr	124,894

The dominance of procedural provisions (Federal Court Act, Civil Procedure Code, Criminal Procedure Code) at the top of the list reflects that every Federal Supreme Court decision must address jurisdictional and procedural questions before reaching the merits. The raw surface-form top 10 (before merging) is dominated by aliases of the Federal Court Act, as expected: `ART.100.ABS.1.LTF` (181,294 mentions), `ART.113.LTF` (111,512), `ART.42.LTF` (90,546), `ART.42.BGG` (68,295), and `ART.66.ABS.1.BGG` (58,926). Counting surface forms separately would double-count the same provision and inflate rank positions; the dataset release includes the alias map as a JSON file so that downstream analyses can canonicalise consistently.

The 284,145 distinct provision identifiers (before alias canonicalisation) span federal statutes, cantonal laws, and international treaties.

4.6 Resolved Legislation Access

The statute references in Section 4.5 are navigable: for each reference, the current text of the cited provision can be retrieved directly. This is accomplished through a two-tier access layer.

Federal legislation — local Fedlex mirror. We download the 80 most-cited Swiss federal laws from Fedlex (`fedlex.data.admin.ch`), the official Swiss federal publication portal, via its SPARQL endpoint in Akoma Ntoso XML format. The laws are parsed and indexed into a local SQLite database (`statutes.db`) containing 39,000 articles in all three official languages (approximately 13,000 articles per language), with 228,559 amendment-reference records for traceability. Included are the core federal codes and procedural acts (OR, ZGB, StGB, StPO, ZPO, BV, SchKG, BGG/LTF, DBG, IPRG, AIG, BVG, KVG, AsylG, and 66 others) at their respective consolidation dates, ranging from 2010-04-01 to 2026-02-01. Lookup by (`statute abbreviation`, `article number`, `language`) returns the current article text in ~1 millisecond.

Cantonal and specialised legislation — LexFind integration. Cantonal laws and federal regulations outside the top 80 are accessed via an on-demand integration with the LexFind.ch legislative search portal, which covers approximately 33,000 federal, cantonal (all 26 cantons), and intercantonal legislative texts. Because the LexFind API is rate-limited and requires a multi-step session protocol (create a search, then paginate), responses are cached locally with tiered TTLs (24 hours for search responses, 30 days for systematic-number resolutions, 7 days for full-text fetches). The current cache contains 696 entries. Cantonal law lookups that are not yet cached typically return within 1–7 seconds.

Practical value for LLM-based legal research. Both Fedlex and LexFind are challenging to access directly from a language model or agent. Fedlex exposes only a SPARQL endpoint and XML downloads; there is no simple HTTP API for “give me the current text of Art. 41 OR in

German.” LexFind requires JavaScript-state management and a search-then-fetch workflow. As a result, language models asked about Swiss statutes without a retrieval layer typically fall back on outdated training data or hallucinate article text outright. The OpenCaseLaw legislation layer resolves this: each statute reference in the corpus graph can be dereferenced to authoritative current text, and the same interface is exposed as a set of tools callable from LLM agents over the Model Context Protocol. A decision’s citation to “Art. 12 BGFA” can therefore be expanded on demand to the full German, French, and Italian text of the article as consolidated on the referenced date, without the LLM needing to navigate Fedlex.

The legislation layer is a companion contribution: it is primarily infrastructure enabling the citation/statute graph to be useful, rather than a standalone research resource. Its construction, caching policy, and known limitations are documented in the source repository.

5. Data Quality and Validation

Scale is insufficient to make a dataset useful; coverage, extraction quality, and resolution accuracy all matter. This section reports what we have validated, what we have not, and where the weaknesses of the current release lie.

5.1 Citation Extraction

Our citation extractor uses regular expressions tuned to Swiss citation conventions. We have not yet performed a formal precision/recall evaluation against a human-annotated gold set. What we can report:

- **Structural checks.** 100% of extracted BGE references parse to valid (volume, division, page) triples. 100% of extracted docket numbers parse to valid federal court case-number patterns. Invalid candidates (e.g., overlapping year ranges) are rejected at extraction time.
- **Resolution rate.** 73.7% of extracted decision-citation references resolve to a target in the corpus. The residual 26.3% is dominated by citations to decisions not published online, foreign jurisdictions, and pre-publication Federal Supreme Court rulings.
- **Known failure modes.** The regular-expression approach misses citations embedded in footnotes with unusual formatting, citations using informal abbreviations (“the *Genf* case”), and citations where the docket number spans a line break in the source PDF. A conservative estimate based on manual spot-checks of 20 decisions is that an additional 2–5% of reference-like strings are missed.

A human-annotated gold set of 100 decisions would be sufficient to report precision/recall and is planned for the next release. We publish the extractor code so that others can evaluate it independently.

5.2 Resolution Accuracy

We spot-checked 50 randomly-sampled resolved citations against the cited decisions. 48/50 (96%) were correctly resolved; 2 were BGE alias ambiguities where the same BGE volume/division/page triple was associated with two candidate decisions and our resolver picked the one with more context matches. We do not report a confidence interval on this small sample; a larger evaluation is planned.

5.3 Deduplication

Our canonical key (normalised court code + docket number + date) is tight by construction, but language variants of the same decision (e.g., a BGer decision published in German and French) can produce two records with different canonical keys. We do not merge language variants; downstream analyses that need to treat them as one should use the `decision_group_id` field (when available) or perform document-level deduplication.

5.4 Field Completeness

See Section 3.3 and Table 5. No field is 100% complete except `docket_number` and `full_text`. Users querying optional fields (`regeste`, `chamber`, `outcome`) should handle missingness explicitly.

5.5 Limitations of This Section

This validation section is intentionally modest. A full data-quality audit would require (a) a human-annotated citation gold set, (b) a manual accuracy check on a larger resolution sample, (c) a deduplication error-rate estimate with independent human review, and (d) a statute-normalisation gold set. We consider these essential next steps and invite contributions.

6. Retrieval Baseline

We release a 100-query multilingual retrieval benchmark to enable reproducible comparisons on Swiss case-law retrieval.

6.1 Benchmark Design

The benchmark comprises 100 queries. Queries were originally tagged with language labels; 74 carry a German tag, 16 a French tag, 7 an Italian tag, and 3 are untagged (their query text is mixed-language or language-ambiguous). Queries span 15 legal domains (social insurance, contract law, criminal law, administrative law, tax law, tenancy, migration, and others). Each query has 1–6 graded relevant decisions (mean 3.04, grade scale 1–3), identified through citation-graph inspection and manual assessment by the author. Relevance judgments are single-annotator; no inter-annotator agreement is reported. The benchmark is released as a JSON file. We consider inter-annotator evaluation a necessary next step and will release an updated benchmark with multi-annotator judgments in a subsequent version.

6.2 Pipeline

The retrieval pipeline uses FTS5 full-text search with BM25 scoring, fused with the citation graph through a handful of domain-specific signals. Pipeline details and scoring weights are available in the source repository. The paper version of the benchmark corresponds to commit `<SHA>` of the repository.

6.3 Results

Table 12. Retrieval baseline on the 100-query benchmark (MRR@10, Hit@1, Recall@10, nDCG@10). All configurations operate on the full 965,141-decision corpus.

Configuration	MRR@10	Hit@1	Recall@10	nDCG@10
Naive BM25	0.320	0.26	0.42	0.34
+ citation-graph signals	0.587	0.50	0.574	0.50
+ LLM-based query parsing	0.611	0.53	0.584	0.51
+ automated weight tuning	0.628	0.54	0.581	0.52
+ LLM-based reranking	0.680	0.60	0.595	–

The dominant source of improvement over the naive BM25 baseline is the addition of citation-graph signals (+0.267 MRR). LLM-based query parsing adds +0.024 MRR (the LLM extracts doctrine names and statute references that are used to augment the lexical query). Automated weight tuning, described briefly in Appendix A, adds a further +0.017. Optional LLM-based reranking on the top 15 results adds another +0.052. We do not report bootstrap confidence intervals in this version; a rigorous comparison with significance testing is planned.

Per-language results and pipeline sensitivity analysis (excluding and including each signal group separately) are reported in the supplementary material released with the benchmark.

7. Availability and Reproducibility

The corpus is updated daily and the main repository advances continuously. To support reproducibility, the version of the corpus, graph, benchmark, and code corresponding to this paper is pinned to fixed identifiers.

Table 13. Pinned artifacts for the April 2026 snapshot.

Artifact	Identifier
Corpus snapshot date	2026-04-10
Decisions in snapshot	965,141
Hugging Face dataset	voilaj/swiss-caselaw
Hugging Face commit	(pinned at publication; see docs/paper/REPRODUCIBILITY.md)
Source code repo	github.com/jonashertner/caselaw-repo-1
Code commit	(pinned at publication)
Benchmark file	benchmarks/search_relevance_golden.json
Reference graph	output/reference_graph.db (re-buildable from corpus + code)
Federal legislation mirror	output/statutes.db (80 laws, 39,000 articles; re-buildable from Fedlex SPARQL)
LexFind cache	output/lexfind_cache.db (tiered TTL; regenerates on demand)
Statute alias map	output/statute_aliases.json (derived from Fedlex)

Users who need the exact numbers in this paper should check out the pinned code commit and use

the pinned Hugging Face commit. The live dataset continues to grow; a direct comparison of future snapshot statistics with the numbers in this paper requires rebuilding against the pinned snapshot.

Removal policy. Once the corpus is mirrored by third parties, removal is not fully reversible. Our governance policy (in the repository) commits to (a) removing specific decisions from the next published Hugging Face snapshot on justified request, (b) honouring court-originated anonymisation requests, and (c) not republishing decisions that have been withdrawn by the originating court. We cannot force third-party mirrors to remove data, and we say so plainly.

8. Ethics and Limitations

8.1 Legal Basis

Published Swiss court decisions are excluded from copyright protection under Art. 5(1)(c) of the Swiss Copyright Act (URG). Federal courts publish decisions pursuant to Art. 27 of the Federal Supreme Court Act (BGG); cantonal publication duties are governed by cantonal procedural codes. The dataset packaging is released under CC0-1.0; all source code is MIT-licensed.

8.2 Privacy

OpenCaseLaw preserves decisions exactly as published by the originating courts, including each court’s anonymisation choices. We do not perform additional anonymisation. Cantonal anonymisation practices vary: some cantons redact party names comprehensively, others publish identifying details for certain case types.

Large-scale aggregation creates privacy risks beyond those of individual court portals (Pilan et al., 2024). The dataset includes a governance and removal policy under which individuals may request removal of specific decisions. Researchers using the corpus for studies involving personal data should conduct their own data-protection assessment.

8.3 Limitations

- **Scope: published decisions only.** The dataset covers only decisions that the originating courts have made publicly available. Unpublished decisions—the majority of Swiss court output—are not included. Publication practice varies by court, era, and case type.
- **Citation extraction is rule-based and not formally evaluated.** Section 5.1 describes structural checks and spot-check observations. A gold-annotated evaluation is not yet provided.
- **Single-annotator benchmark.** The retrieval benchmark reflects one author’s relevance judgments. Inter-annotator agreement is not reported.
- **Historical text quality varies.** Pre-1990 material may contain OCR artifacts. Approximately 5,000 scanned-PDF decisions could benefit from improved OCR.
- **Commercial comparisons are approximate.** Decision counts for Swisslex and Weblaw in Table 1 are estimates; exact figures are not publicly audited.
- **Statute alias canonicalisation is manual.** The alias map (LTF = BGG, ZPO = CPC, etc.) is hand-maintained. Additional language variants may exist.
- **Cross-language findings are descriptive, not causal.** Section 4.3 explicitly does not attempt a controlled analysis of legal authority flow.

Appendix A: Automated Weight Tuning (Meta-Harness)

The retrieval pipeline exposes 55+ numeric scoring parameters (BM25 column weights, citation-graph signal strengths, fusion weights). Manually tuning these is tedious. We implemented a Meta-Harness-style optimizer (Lee et al., 2026) that treats the full scoring configuration as a search space. At each iteration, the optimizer evaluates the current configuration on the benchmark, generates per-query execution traces showing which candidates were retrieved and at what rank, and asks an LLM proposer to suggest new weights targeting the weakest query categories. The optimizer converged within 2–3 iterations to the configuration reported in Section 6. The optimized weights add no runtime cost; they are constants in the scoring function. This appendix is included for transparency about the origin of the Table 12 numbers. A detailed description of the optimizer and its full configuration space is available in the source repository.

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